CE444: Geotechnical Design

Homework Assignment #6

March 13th, 2026

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From: Brandon Auyeung

Problem 1.

GIVEN:

A single pile foundation is being considered to support a structure with a weight of 1.8 MN. The site investigation program includes one CPT performed at the center of the site (attached file). The type of piles readily available at the site is driven RC concrete piles with a circular cross section and a diameter of 2 feet (0.61 meter). Assuming a factor of safety (FS) of 4 find the required length of pile knowing that at least 12 meters of pile is needed to provide stability against lateral loads. Follow the following steps and assumptions:

Analyze CPT data using the spreadsheet that you developed in Homework 3. Assume water table is at ground surface. Total unit weight is 17 kN/m3. CPT tip resistance is already area-corrected. Calculate soil behavior index (I_c) using Robertson (1990). Classify the soil type to either Sand or Clay based on Ic threshold of 2.6. Estimate effective friction angle in sand using correlations by Robertson and Campanella (1983). Estimate undrained shear strength in clay using Nk factor of 15.

Develop idealized soil profile and fill out table below. Develop "design" soil properties by averaging friction angle and shear strength in each soil layer.

FIND:

Calculate skin friction (q_s) and end bearing (q_p) for each data point in the CPT file based on whether that data point corresponds to Clay or Sand. Get an estimate of friction angle between concrete pile and soil (δ) using AASHTO LRFD (2014) Table 3.11.5.3-1 (attached). Use your engineering judgment in selecting the right value from table. Use the "design" soil properties in your calculation. Calculate resistance Q_s and Q_p profiles. Find allowable resistance (Q_{all}) and determine required length of pile.

In your submittal, include: table below, individual figures for skin friction (q_s) profile (versus depth), q_p profile, Q_{all} profile (in all figures show the soil layers with straight lines and label soil layers as Sand or Clay), total required pile length, and comment on the comparison of q_s and q_p in sand versus clay layers.

Sublayer	Type (Sand/Clay)	Top Depth (m)	Bottom Depth (m)	Avg ' (Deg)	N*q	Avg Cu (kPa)	K	α
1								
2								
3								
4								
5								
6								

SOLUTION:

CPT data was analyzed to generate the following CPT profiles depicted in **Figures 2 and 3**. The average ϕ' and S_u were collected from each sublayer. The value of N_q^* was obtained using the figure generated by Meyerhof in **Figure 1**.

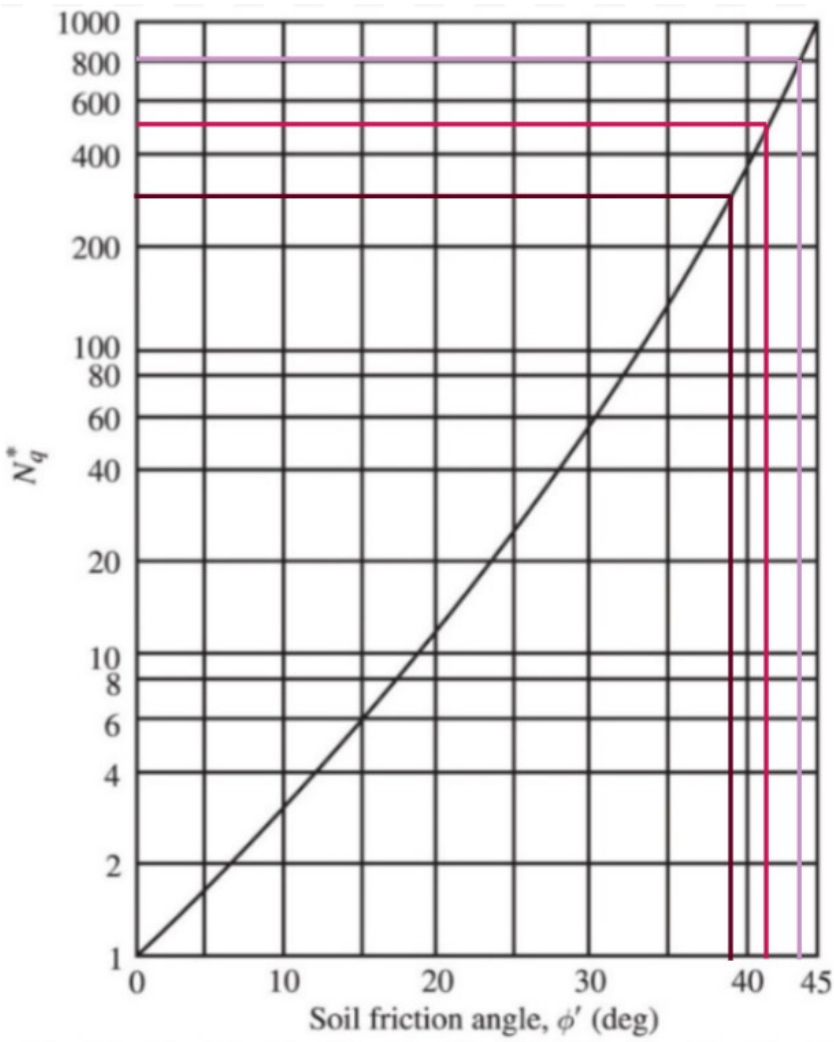


Figure 1. Selection of N_q^* based on extracted ϕ'.

The value of K was determined using the below equation.

$$K = 1.8(1 - \sin\phi') \qquad \textit{Driven Pile}$$

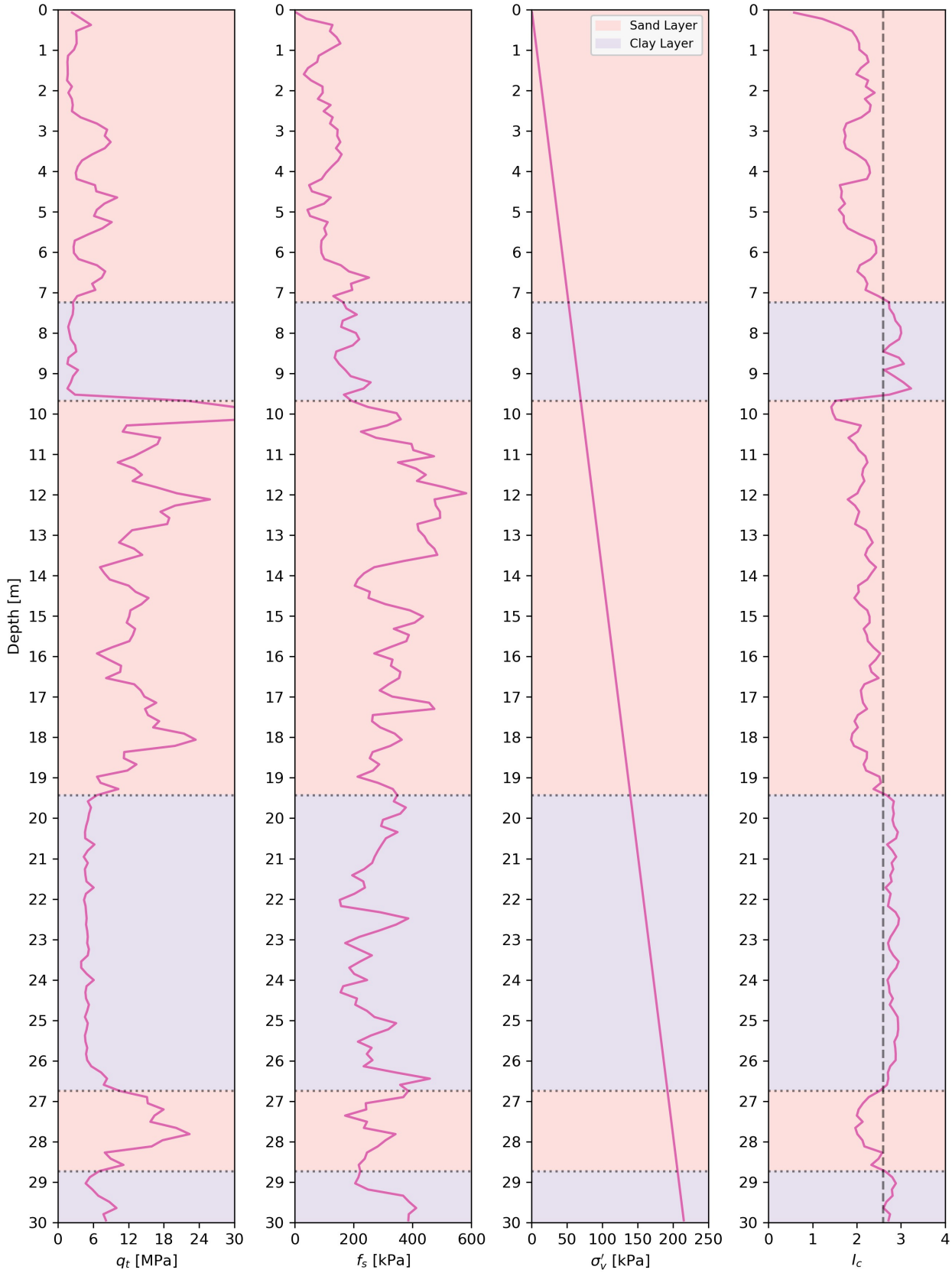


Figure 2. Tip resistance, sleeve friction, vertical effective stress, and I_c values of the profile.

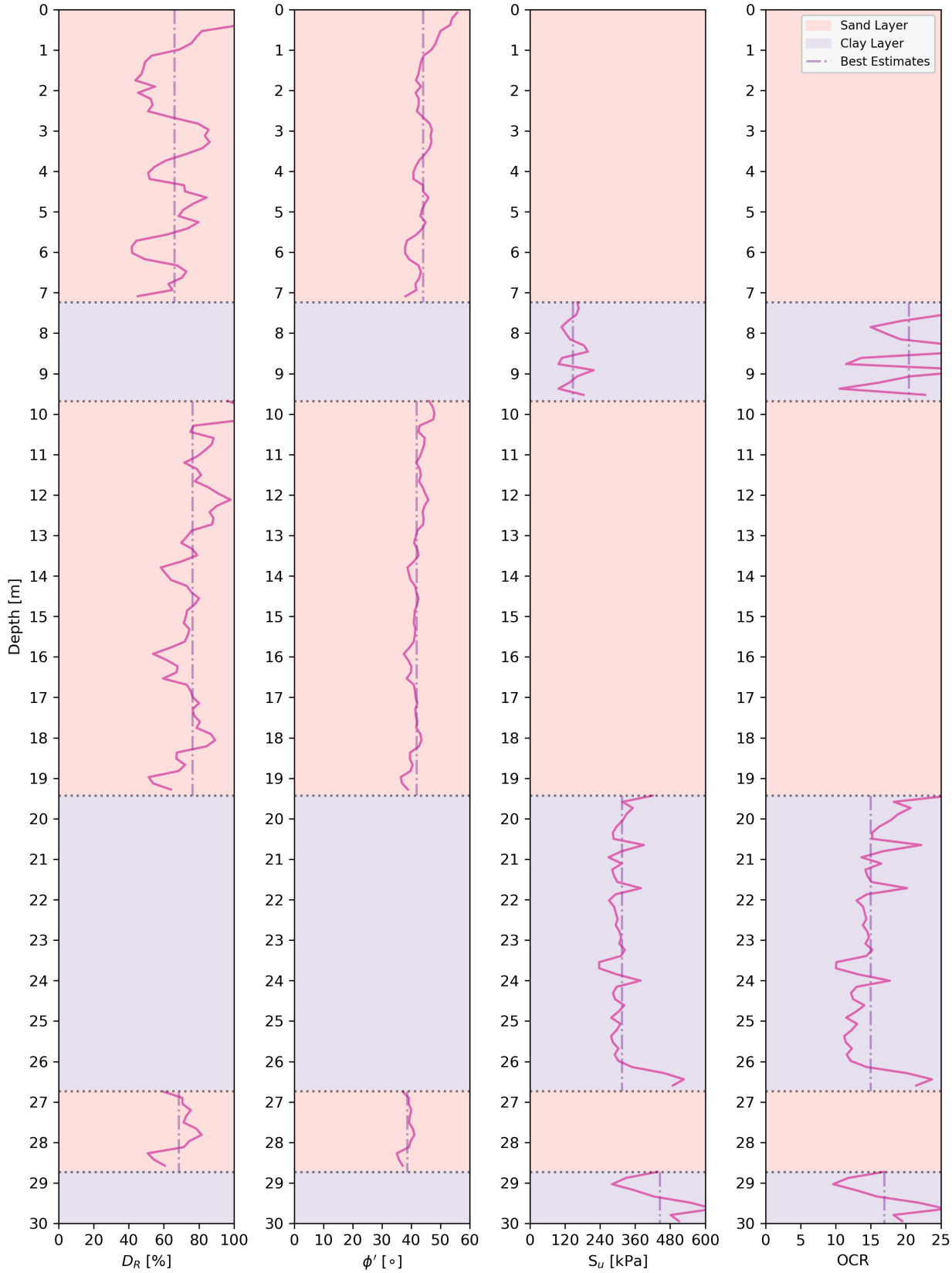
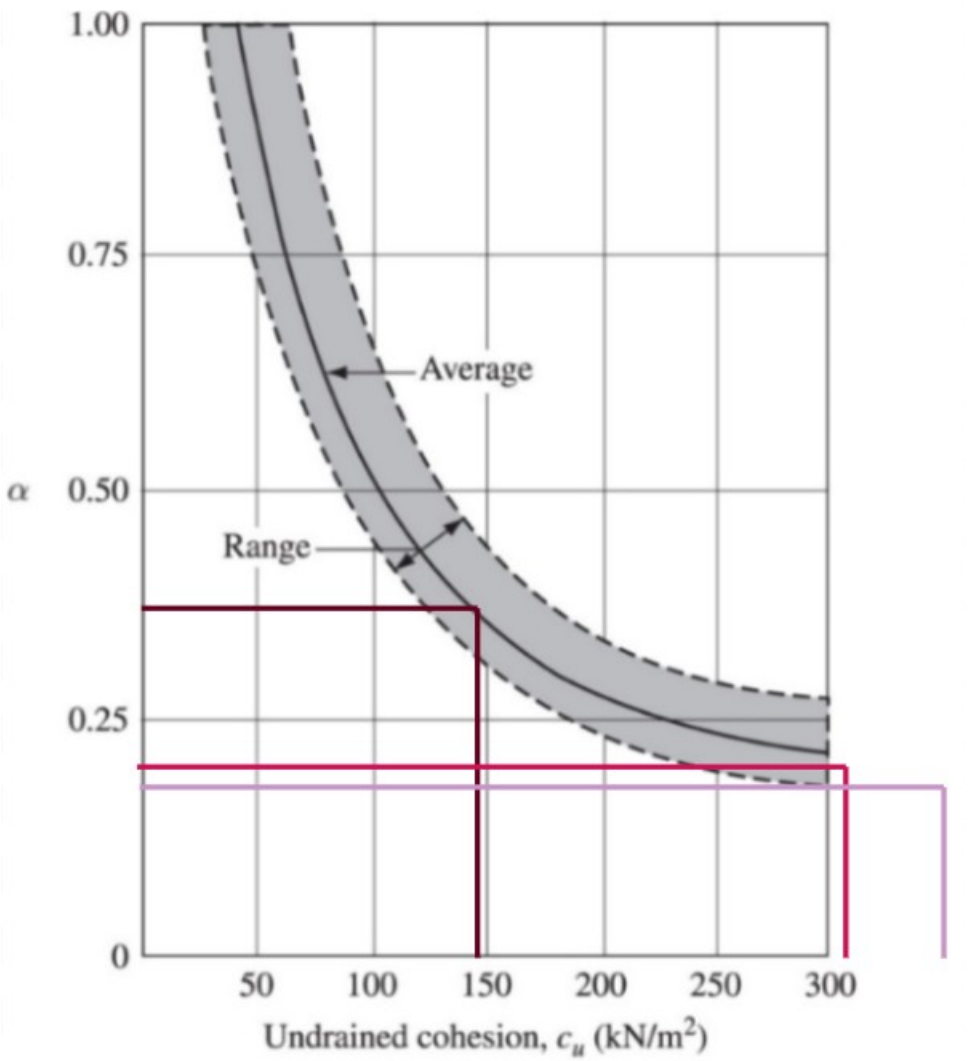


Figure 3. Relative density, friction angle, S_U, and OCR of the soil profile.

The value of α was obtained from the figure generated by Das B.M. as shown in **Figure 4.** Extrapolation was required due to S_u values extracted being higher than graph axis limits.



Das B.M. (2015) Geotechnical Engineering, 4th Edition.

Figure 4. Selection of α based on extracted S_u with extrapolation.

Sublayer	Soil Type	Top Depth [m]	Bottom Depth [m]	AVG(ϕ') [○]	N_q^*	AVG(S_u) [kPa]	K	α
1	Sand	0	7.2	44.1	800	n/a	0.55	n/a
2	Clay	7.2	9.7	n/a	n/a	148	n/a	0.38
3	Sand	9.7	19.4	41.9	500	n/a	0.60	n/a
4	Clay	19.4	26.7	n/a	n/a	316	n/a	0.20
5	Sand	26.7	28.7	38.7	300	n/a	0.67	n/a
6	Clay	28.7	30.0	n/a	n/a	445	n/a	0.18

Table 1. Design values for soil profile.

Using these design values, the values of q_p and q_s can be obtained.

SAND

$$q_p = N_q^* \sigma_v' \leq 50 \tan\phi' N_q^*$$

$$q_s = \sigma_v' K \tan\delta \text{ where } \delta = 17^\circ$$

δ obtained from AASHTO LRFD (2014) Table 3.11.5.3-1

Interface Materials	Friction Angle, δ (degrees)	Coefficient of Friction, tan δ (dim.)
Mass concrete on the following foundation materials:		
• Clean sound rock	35	0.70
• Clean gravel, gravel-sand mixtures, coarse sand	29 to 31	0.55 to 0.60
• Clean fine to medium sand, silty medium to coarse sand, silty or clayey gravel	24 to 29	0.45 to 0.55
• Clean fine sand, silty or clayey fine to medium sand	19 to 24	0.34 to 0.45
• Fine sandy silt, nonplastic silt	17 to 19	0.31 to 0.34
• Very stiff and hard residual or preconsolidated clay	22 to 26	0.40 to 0.49
• Medium stiff and stiff clay and silty clay	17 to 19	0.31 to 0.34
Masonry on foundation materials has same friction factors.		
Steel sheet piles against the following soils:		
• Clean gravel, gravel-sand mixtures, well-graded rock fill with spalls	22	0.40
• Clean sand, silty sand-gravel mixture, single-size hard rock fill	17	0.31
• Silty sand, gravel or sand mixed with silt or clay	14	0.25
• Fine sandy silt, nonplastic silt	11	0.19
Formed or precast concrete or concrete sheet piling against the following soils:		
• Clean gravel, gravel-sand mixture, well-graded rock fill with spalls	22 to 26	0.40 to 0.49
• Clean sand, silty sand-gravel mixture, single-size hard rock fill	17 to 22	0.31 to 0.40
• Silty sand, gravel or sand mixed with silt or clay	17	0.31
• Fine sandy silt, nonplastic silt	14	0.25
Various structural materials:		
• Masonry on masonry, igneous and metamorphic rocks:		
○ dressed soft rock on dressed soft rock	35	0.70
○ dressed hard rock on dressed soft rock	33	0.65
○ dressed hard rock on dressed hard rock	29	0.55
• Masonry on wood in direction of cross grain	26	0.49
• Steel on steel at sheet pile interlocks	17	0.31

Table 2. AASHTO LRFD (2014) Table 3.11.5.3-1.

CLAY

$$q_p = 9S_u$$

$$q_s = \alpha S_u$$

The value of Q_p and Q_s were obtained as follows.

$$Q_p = q_p A_{base} \text{ where } A_{base} = \frac{\pi}{4}(0.61 \text{ m})^2 = 0.2920 \text{ m}^2$$

$$\Delta Q_s = \text{AVG}(q_s[i], q_s[i-1]) A_s \text{ where } i \equiv \text{current value in list, } A_s = \pi(0.61 \text{ m})(z[i+1] - z[i]) = 0.2922 \text{ m}^2$$

$$Q_s = \sum_i^{30 \text{ m}} \Delta Q_s$$

The values of Q_L and Q_{all} were obtained as follows.

$$Q_L = Q_p + Q_s$$

$$Q_{all} = \frac{Q_L}{\text{F.S.}} \text{ where F.S.} = 4$$

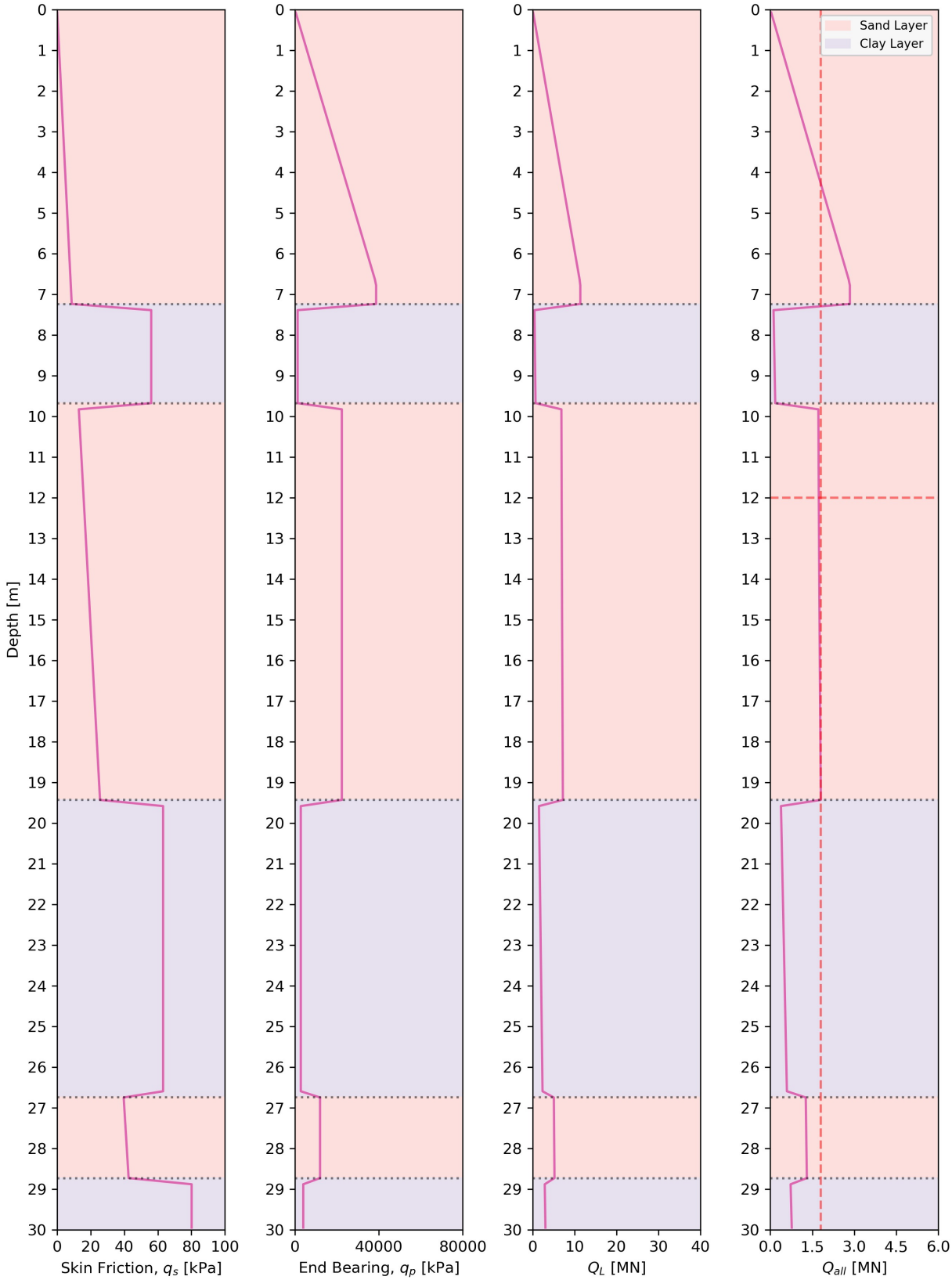


Figure 5. q_s, q_p, Q_L and Q_{all} profiles.

The required pile length must be longer than 12 meters and must have an allowable capacity of 1.8 MN. Therefore, the pile length must be 19 meters.

In the sand layer, the end bearing pressure is much greater than the skin friction pressure. This is because sand's cohesionless properties do not provide much sticking friction to the pile, while the stress-dependent behavior of sand allows intergranular resistance to mobilize.

In the clay layer, the skin friction pressure is much greater than the end bearing pressure. This is because clay has cohesion, allowing it to stick to the pile, while the softer clay cannot provide high end bearing capacity.

Problem 2.

GIVEN:

In this problem we will practice how to perform lateral load analysis on piles using the Beam-on-Nonlinear-Winkler Foundation (BNWF) method, also known as the p-y approach. There are many commercial and non-commercial software packages available for lateral load analysis of piles, e.g. Lpile by Ensoft, PB-MultiPier, DFSAP, RSPile by Rocscience, and OpenSees, to name a few. In this problem we will use Lpile Student Version to solve the problem, but any other lateral pile analysis software can be used too.

FIND:

Follow instructions to run multiple pile tests. Answer the following based on each simulation.

TEST ONE

The four plots of deflection, soil reaction, shear, and moment with depth. You can take a snapshot and include it in your report or use the output file to extract digital data and plot them in excel. If you are taking a snapshot, your plots in your report should be legible. Fill out the following table and include in your report:

Boundary conditions at pile head	Maximum lateral deflection (m)	Maximum shear force (kN)	Maximum bending moment (kN-m)
V = M = Axial force =			

TEST TWO

From the Graphics menu open the Load vs. Top Deflection window. This plot is called the pile-head force-displacement or the pushover curve. Include this figure in your report. Calculate and report the pile lateral stiffness (in kN/m), which is the slope of the initial portion of the pushover curve.

TEST THREE

The four plots of deflection, soil reaction, shear, and moment with depth. You can take a snapshot and include it in your report or use the output file to extract digital data and plot them in excel. If you are taking a snapshot, your plots in your report should be legible. Fill out the following table and include in your report:

Boundary conditions at pile head	Maximum lateral deflection (m)	Maximum shear force (kN)	Maximum bending moment (kN-m)
V = M = Axial force =			

SOLUTION:

TEST ONE

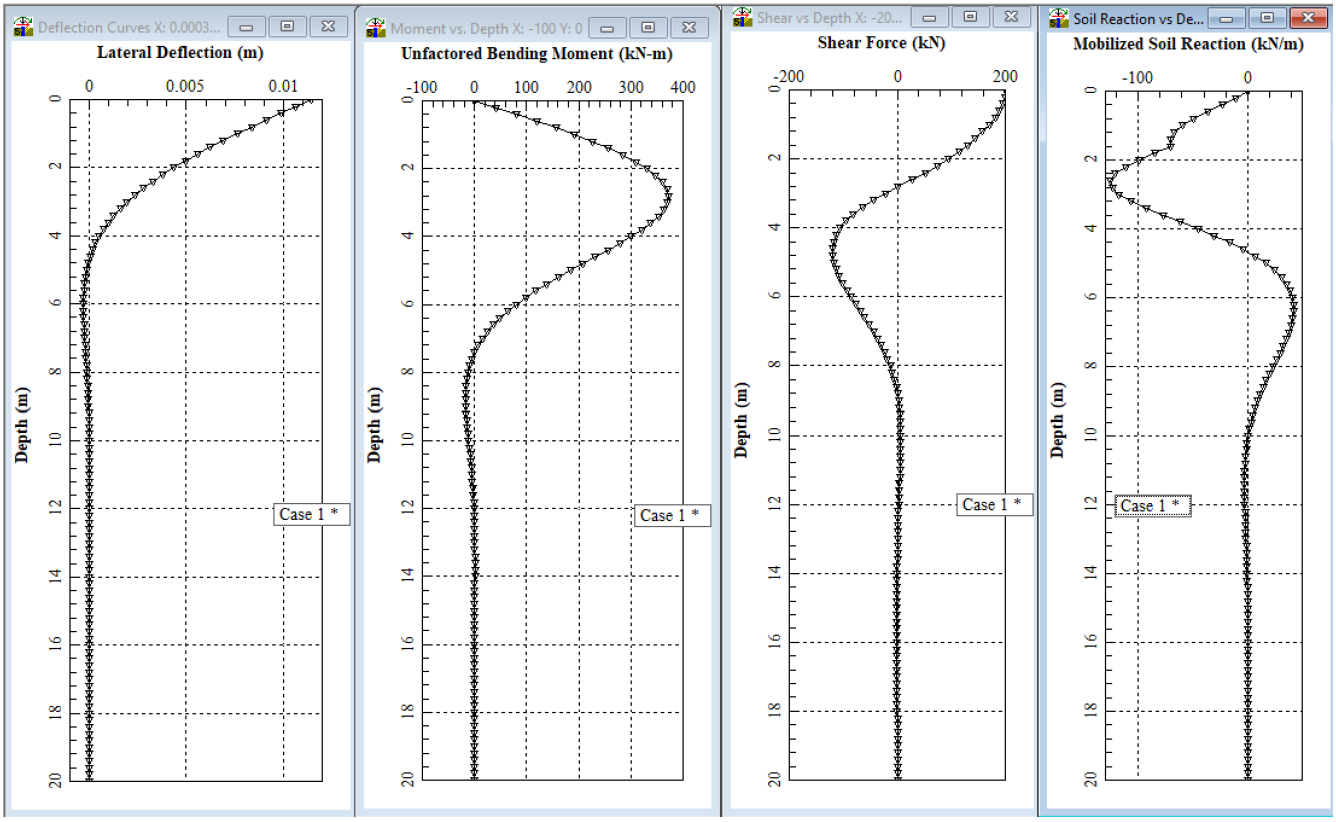


Figure 6. Curves generated from *Test One* on sand.

Boundary conditions at pile head	Maximum lateral deflection (m)	Maximum shear force (kN)	Maximum bending moment (kN-m)
V = 200 kN M = 0 Axial force = 500 kN	0.11417	200	372.2

Table 3. Results from *Test One*.

Test Two

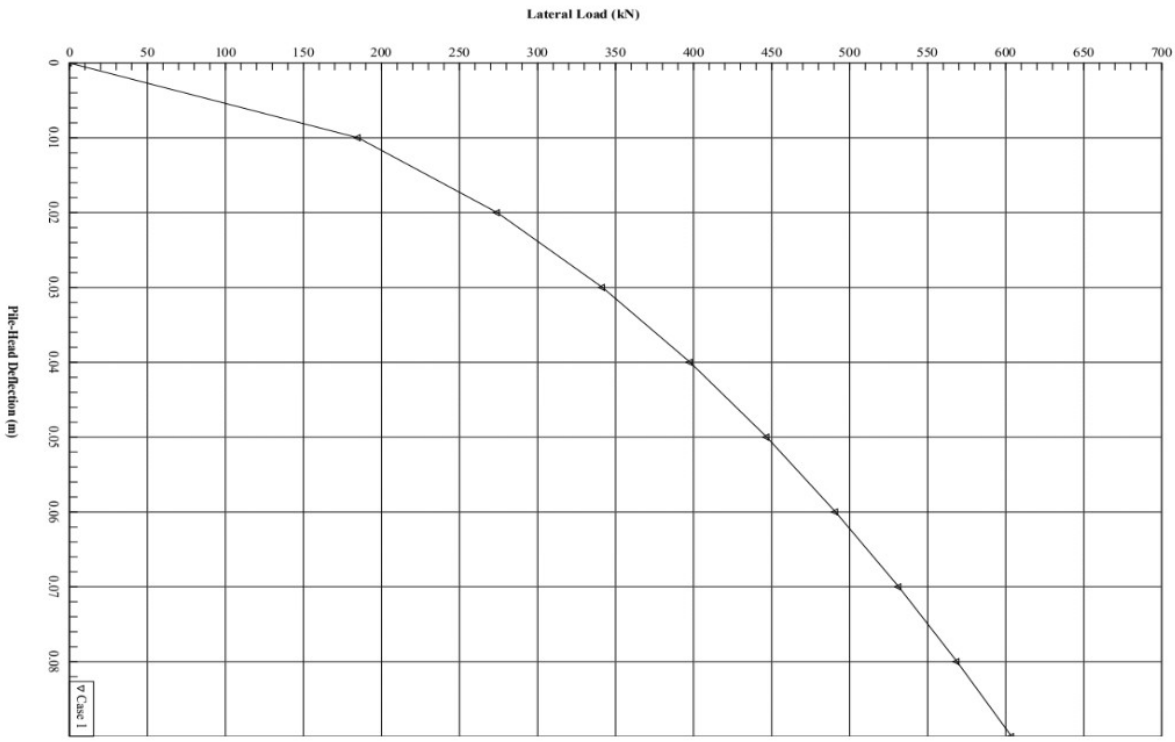


Figure 7. Pushover curve generated from *Test Two*.

$$k \equiv \text{initial slope} = \frac{185.013 \text{ kN} - 0}{0.01 \text{ m} - 0} = 18501.3 \text{ kN/m}$$

Test Three

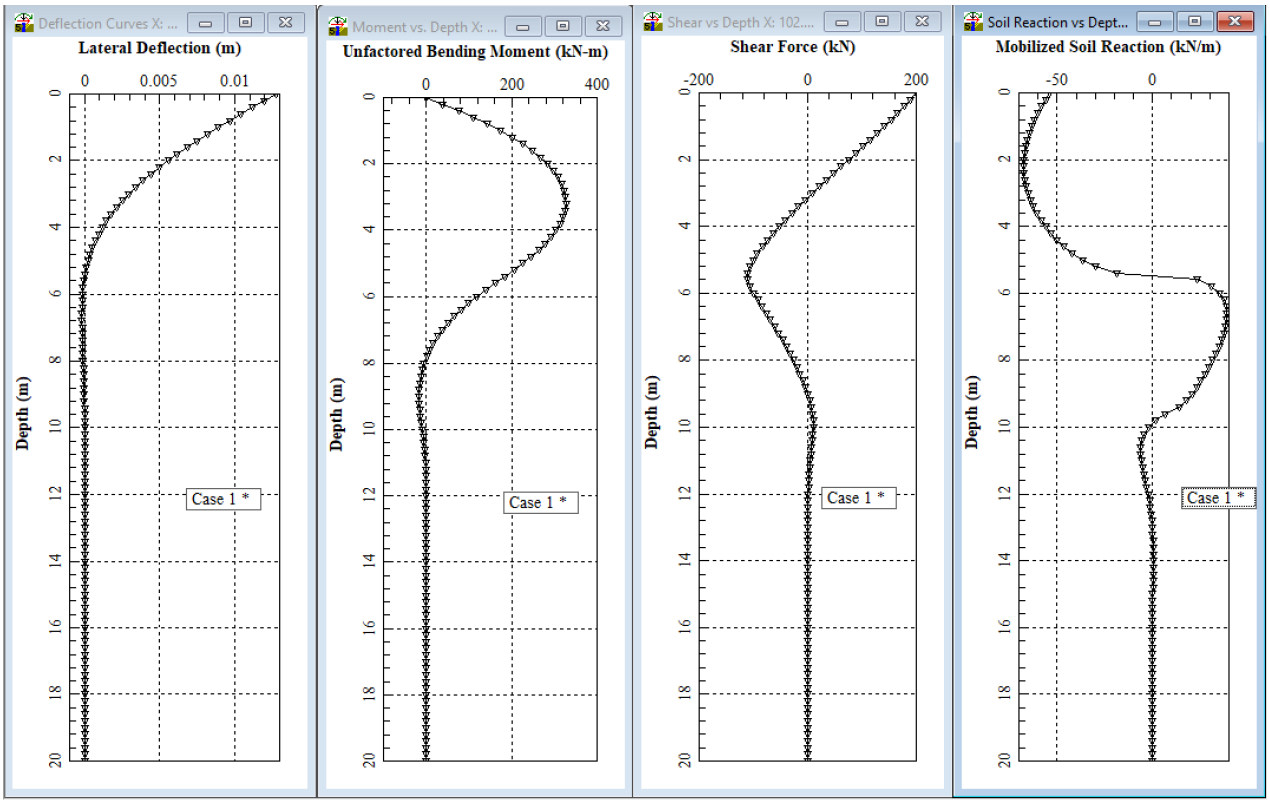


Figure 8. Curves generated from *Test Three* on clay.

Boundary conditions at pile head	Maximum lateral deflection (m)	Maximum shear force (kN)	Maximum bending moment (kN-m)
V = 200 kN M = 0 Axial force = 500 kN	0.12758	200	372.3

Table 3. Results from *Test Three*.